

ANATOMY-AWARE CROSS-SPEAKER ADAPTATION OF COMPLETE VOCAL-TRACT ACOUSTIC-TO-ARTICULATORY INVERSION

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ABSTRACT

Cross-speaker acoustic-to-articulatory inversion requires accounting for anatomical differences between speakers. We propose a geometric adaptation framework that uses anatomical landmarks, primarily on vertebrae and dental structures, to transfer predictions from a fixed inversion model to unseen speakers. An affine transformation followed by thin-plate spline (TPS) deformation maps the predicted contours of 10 vocal-tract structures into each target speaker's geometry without retraining. Landmarks are identified in one selected /u/ frame per speaker as a common phonetic reference without assuming identical articulatory configurations across speakers, and the resulting mapping is reused across recordings. We train the model on a single-speaker rt-MRI database and evaluate adaptation on eight speakers from a separate multi-speaker rt-MRI database. We compare affine and TPS configurations using 12 or 14 landmarks. Affine12+TPS14 achieves the lowest mean point-to-closest-point error of 3.19 mm. These results support the combined value of anatomical landmark information and nonrigid alignment.

Index Terms— Acoustic-to-articulatory inversion, speaker adaptation, anatomical landmarks, thin-plate splines

1. INTRODUCTION

Acoustic-to-articulatory inversion (AAI) estimates speech articulation from acoustics. Generalization to unseen speakers remains challenging because vocal-tract anatomy and articulatory strategies vary across individuals [1], while paired acoustic-articulatory recordings are costly to collect [2]. Speaker-independent models and adaptation methods address this challenge using vocal tract length normalization [3] and self-supervised speech representations [4].

The articulatory representation also determines the scope of inversion. Electromagnetic articulography (EMA) tracks sparse sensors on easily accessible articulators, whereas real-time magnetic resonance imaging (rt-MRI) captures midsagittal anatomy, including pharyngeal and laryngeal regions. Previous studies have reconstructed rt-MRI images from speech [5], while automatic segmentation has enabled contour-based

representations [6]. Azzouz et al. demonstrated acoustic reconstruction of contours spanning the vocal tract from the glottis to the lips [7]. Transferring these predictions to unseen speakers, however, requires addressing geometric mismatch.

Geometric registration has been investigated through affine mappings between EMA-derived articulatory systems [8]. Wei and Dang used grid-defined landmarks and thin-plate spline (TPS) warping to normalize multi-speaker EMA data to a shared template [9]. Anatomical information has also been incorporated through joint estimation of anatomy and articulation [10] and speech-conditioned rt-MRI generation using static MRI [11]. Here, we instead use anatomical landmarks to transfer complete MRI-derived contour predictions from a fixed AAI model to unseen speaker geometries using only single-frame anatomical calibration, without model retraining. We examine how landmark coverage and nonlinear alignment contribute under this constraint.

We estimate an affine-TPS reference-to-target mapping from primarily vertebrae and dental landmarks identified in one /u/ frame per speaker under comparable phonetic conditions. TPS provides smooth nonrigid deformation constrained by landmark correspondences. Each mapping is reused across recordings without updating the inversion model. We train on a single-speaker rt-MRI database and evaluate on eight speakers from a separate multi-speaker database across 10 structures: the upper and lower lips, tongue, soft-palate midline, pharyngeal wall, epiglottis, arytenoid cartilage, vocal folds, and upper and lower incisors. Comparing affine and affine-TPS configurations with 12 or 14 landmarks assesses the contributions of additional correspondences and nonlinear deformation.

2. DATASETS

We use two rt-MRI corpora from the ArtSpeech databases (ASD) acquired at the Centre Hospitalier Régional Universitaire de Nancy: ASD2 for training the reference inversion model and ASD1 for evaluating cross-speaker adaptation. ASD1 comprises five male and five female native French speakers, each contributing approximately 15 minutes of

speech across 77 sentences, distributed over 16 sessions (15 usable for P4). The evaluated ASD1 data contain approximately 130,000 non-silent frames [12, 13]. We evaluate P1 and P3-P9, excluding P2 because insufficient laryngeal visibility prevents reliable evaluation of the retained structures.

ASD2 contains approximately 3.5 hours of speech from a single native French female speaker, with about 2100 sentences across 153 acquisitions [14]. Her presence as P10 in ASD1 provides a shared speaker identity across corpora [13].

Following the contour-tracking framework in [13], each of these 10 structures is represented by 50 two-dimensional points, yielding 1000 output coordinates per frame.

3. METHODS

3.1. Acoustic-to-articulatory inversion model

We use the backbone introduced by Azzouz et al. [7], comprising two fully connected layers followed by two bidirectional long short-term memory (Bi-LSTM) layers, each with 300 units.

ASD2 acquisitions are rigidly registered using the static-head registration procedure of Azzouz et al. [14]. Reference landmarks are identified after registration, in the same coordinate system as the training contours and denormalized predictions. Previous work uses local moving normalization to compensate for slow variations in articulatory coordinates [2, 7]. For geometric adaptation, we instead use global coordinate-wise statistics computed exclusively from the reference training data:

$$\begin{aligned}\tilde{\mathbf{y}}_t &= (\mathbf{y}_t - \boldsymbol{\mu}_{\text{train}}) \oslash \boldsymbol{\sigma}_{\text{train}}, \\ \hat{\mathbf{y}}_t^R &= \tilde{\mathbf{y}}_t \odot \boldsymbol{\sigma}_{\text{train}} + \boldsymbol{\mu}_{\text{train}},\end{aligned}\quad (1)$$

where $\oslash$ and $\odot$ denote element-wise division and multiplication, respectively. The same training statistics are reused during validation and inference to recover reference-space contour coordinates. No target-speaker contour sequences are required to estimate the articulatory normalization statistics.

3.2. Geometric adaptation

For each target speaker s , anatomical landmark correspondences define a fixed mapping $G_{R \rightarrow s}$ from predicted reference-space contours to the target geometry, reused across recordings.

3.2.1. Anatomical landmarks

As shown in Fig. 1, the 12-landmark configuration comprises five points sampled along the inferior boundary of the upper-incisor and hard-palate contour ($I1$ - $I5$), the centers of the cervical vertebrae ($C1$ - $C6$), and a posterior landmark $P1$. The bony and dental landmarks provide anatomical anchors, while $P1$ supplies an additional correspondence on the pharyngeal wall.

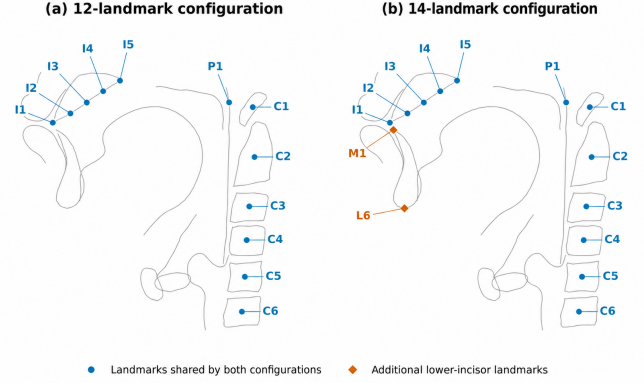


Fig. 1. Anatomical landmark configurations used for geometric adaptation. The 12-landmark set (in blue) is shown together with the two additional lower-incisor landmarks, M1 and L6 (in orange), used in the 14-landmark configuration.

The landmark $P1$ is defined as the intersection of the line through $I5$ and the center of $C1$ with the pharyngeal-wall contour. The 14-landmark configuration additionally includes $M1$ and $L6$, defined as the superior and inferior extrema, respectively, of the lower-incisor contour in the midsagittal image. These landmarks extend coverage to the lower anterior oral cavity and constrain mandibular alignment. These landmark definitions are applied to the calibration frames described in Section 3.4.

3.2.2. Geometric transformation

In the proposed configuration, the 12-landmark set first estimates a global affine alignment accounting for translation, rotation, scaling, and shear, after which $M1$ and $L6$ provide two additional constraints for the subsequent nonrigid deformation:

$$(\mathbf{A}_{R \rightarrow s}, \mathbf{b}_{R \rightarrow s}) = \arg \min_{\mathbf{A}, \mathbf{b}} \sum_{k=1}^{12} \|\mathbf{A} \mathbf{l}_{R,k} + \mathbf{b} - \mathbf{l}_{s,k}\|_2^2. \quad (2)$$

where $\mathbf{l}_{R,k}$ and $\mathbf{l}_{s,k}$ are the corresponding two-dimensional landmark coordinates, $\mathbf{A}_{R \rightarrow s} \in \mathbb{R}^{2 \times 2}$, and $\mathbf{b}_{R \rightarrow s} \in \mathbb{R}^{2 \times 1}$. The affine transformation acts on a reference point $\mathbf{p} \in \mathbb{R}^{2 \times 1}$ as $T_{R \rightarrow s}^{\text{Affine}}(\mathbf{p}) = \mathbf{A}_{R \rightarrow s} \mathbf{p} + \mathbf{b}_{R \rightarrow s}$.

The affine transformation is then applied to all 14 reference landmarks, including $M1$ and $L6$, giving $\mathbf{c}_{s,k} = T_{R \rightarrow s}^{\text{Affine}}(\mathbf{l}_{R,k})$. TPS is fitted from these affine-transformed landmarks to their target correspondences $\mathbf{l}_{s,k}$:

$$T_{R \rightarrow s}^{\text{TPS}}(\mathbf{q}) = \mathbf{a}_{s,0} + \mathbf{B}_s \mathbf{q} + \sum_{k=1}^{14} \mathbf{w}_{s,k} U(\|\mathbf{q} - \mathbf{c}_{s,k}\|_2). \quad (3)$$

where $\mathbf{q}$, $\mathbf{a}_{s,0}$, $\mathbf{c}_{s,k}$, $\mathbf{w}_{s,k} \in \mathbb{R}^{2 \times 1}$, $\mathbf{B}_s \in \mathbb{R}^{2 \times 2}$, and $U(r) = r^2 \log r$ for $r > 0$, with $U(0) = 0$ [15]. TPS is fitted by exact

interpolation with zero smoothing and a first-degree polynomial term. The complete reference-to-target mapping is

$$G_{R \rightarrow s}(\mathbf{p}) = T_{R \rightarrow s}^{\text{TPS}}(T_{R \rightarrow s}^{\text{Affine}}(\mathbf{p})). \quad (4)$$

This mapping is applied point-wise to the predicted reference-space contours.

3.3. Evaluation

Following Ribeiro [13], we evaluate contour reconstruction using the mean point-to-closest-point distance ($\text{P2CP}_{\text{mean}}$). Unlike point-wise RMSE, this metric does not require matching point indices across predicted and target contours, making it suitable for cross-speaker geometric evaluation.

Before evaluation, predicted and target contours are independently regularized using quadratic B-splines with 20 subintervals and a regularization parameter of 0.1, then resampled to $N = 50$ points. For contours $\mathbf{u}$ and $\mathbf{v}$,

$$\text{P2CP}_{\text{mean}}(\mathbf{u}, \mathbf{v}) = \frac{1}{2N} \sum_{i=1}^N \left[\min_j d(\mathbf{u}_i, \mathbf{v}_j) + \min_j d(\mathbf{v}_i, \mathbf{u}_j) \right], \quad (5)$$

where d denotes the Euclidean distance. Distances are reported in millimeters. For each structure, mean errors are first computed over frames within each session, then averaged equally across sessions and speakers. Aggregate means give equal weight to the selected structures.

Standard deviations (SDs) describe pooled frame-level errors for individual structures and pooled frame structure errors for speaker-wise results. Means retain the equal-weight averaging described above.

3.4. Experimental setup

The reference inversion model is trained on ASD2 using the partitions and training settings of Azzouz et al. [14], with the 10-structure output and global normalization described in Section 3.1. Acoustic inputs comprise 13 MFCCs and their first- and second-order derivatives (39 dimensions), extracted with a 25-ms window and a 10-ms hop. MRI contours sampled every 20 ms are aligned to the 10-ms acoustic grid by inserting coordinate-wise averages of consecutive contours [7]. The lowest-validation-RMSE checkpoint is kept fixed throughout adaptation.

Calibration uses one /u/ frame from ‘pour’ in ASD2 and ‘pourri’ for each ASD1 target speaker. The vowel /u/ was selected because it provides a relatively constrained articulatory configuration [16], without assuming identical articulation across speakers. Although the reference speaker also produces ‘pourri’ in ASD1 (P10), we use the registered ASD2 frame to match the geometry of the model predictions. The reference speaker’s recordings differ in head posture between corpora, which can alter vocal-tract shape, particularly in the

pharyngeal region [17]. The words share the immediate /p/-/u/-/b/ context, although their broader contexts differ.

Frames nearest the /u/ midpoint are selected from manually corrected TextGrid phonetic segmentation, independently of reconstruction errors. One mapping is estimated per target speaker and reused across recordings; reference frames are excluded from evaluation.

Raw denotes unadapted predictions. A12 (Affine12) and A14 (Affine14) use 12 and 14 landmarks for affine alignment, respectively. A12+T12 (Affine12+TPS12) and A12+T14 (Affine12+TPS14) apply affine alignment with 12 landmarks, followed by TPS fitted using 12 and 14 affine-transformed reference landmarks, respectively. The 14-landmark set adds M1 and L6. A14 is the affine baseline using the same overall landmark set as A12+T14. All configurations share predictions, calibration frames, evaluation data, and common-landmark definitions.

$\text{P2CP}_{\text{mean}}$ is reported for the 10 evaluated structures (Full10) and for a seven-structure subset excluding the arytenoid cartilage, epiglottis, and vocal folds (Full7). Full7 assesses whether improvements extend beyond the three laryngeal structures. A12+T14 is compared with A14 using two-sided paired t -tests at an unadjusted significance level of 0.05. Paired observations are session means for within-speaker tests and session-balanced speaker means for articulator-wise and Full10/Full7 tests ($n = 8$ speakers).

4. RESULTS

The ASD2-trained model achieves a mean $\text{P2CP}_{\text{mean}}$ of 1.40 mm across the 10 evaluated structures (Table 1). On target speakers, the unadapted Full10 error reaches 8.87 mm, indicating substantial geometric mismatch (Table 3).

A12+T14 achieves the lowest Full10 error of 3.19 mm, improving over A12 (4.10 mm), A14 (3.69 mm), and A12+T12 (3.75 mm) (Table 2). The reduction relative to the main affine baseline, A14, is approximately 13.6% and statistically significant ($p = 0.025$). Full7 error also decreases significantly, from 3.17 to 2.71 mm ($p < 0.05$).

A12+T14 yields the lowest error for six of the 10 structures and improves over A14 for eight. The largest absolute reductions occur for the vocal folds (6.07 to 4.86 mm), pharyngeal wall (3.02 to 2.19 mm), and arytenoid cartilage (4.34 to 3.64 mm). Tongue error decreases from 4.06 to 3.62 mm. At the structure level, only the arytenoid cartilage and lower incisor show statistically significant reductions ($p < 0.05$). Epiglottis and soft-palate errors increase slightly, indicating uneven benefits across structures.

All adaptation configurations outperform Raw for every target speaker. A12+T14 performs best for seven of eight speakers, with significant improvements over A14 for these seven ($p < 0.05$). For P4, however, error increases significantly from 2.95 to 3.17 mm. A12+T14 speaker-wise errors range from 2.46 to 3.95 mm.

Table 1. ASD2 model trained on ten articulators with global normalization. P2CP_{mean} in mm.

Articulator	Mean $\pm$ SD	Median
Arytenoid cartilage	1.53 $\pm$ 0.76	1.38
Epiglottis	1.71 $\pm$ 1.18	1.39
Lower lip	1.29 $\pm$ 0.64	1.17
Pharyngeal wall	0.96 $\pm$ 0.44	0.85
Soft-palate midline	1.01 $\pm$ 0.55	0.88
Tongue	1.99 $\pm$ 0.74	1.84
Upper lip	1.37 $\pm$ 0.76	1.18
Vocal folds	1.55 $\pm$ 0.90	1.37
Lower incisor	1.29 $\pm$ 0.60	1.19
Upper incisor	1.32 $\pm$ 0.78	1.14
Mean	1.40 $\pm$ 0.82	1.22

Table 2. Articulator-wise P2CP_{mean} (mm). A14 is the main affine baseline; A12+T14 is the proposed configuration.

Articulator	A12	A14	A12+T12	A12+T14
Arytenoid	5.09 $\pm$ 2.33	4.34 $\pm$ 2.16	4.52 $\pm$ 2.27	3.64* $\pm$ 1.89
Epiglottis	4.94 $\pm$ 2.74	4.29 $\pm$ 2.40	5.00 $\pm$ 2.76	4.39 $\pm$ 2.43
Lower lip	3.26 $\pm$ 1.50	2.67 $\pm$ 1.06	2.91 $\pm$ 1.20	2.53 $\pm$ 1.11
Pharynx	2.87 $\pm$ 1.79	3.02 $\pm$ 1.91	2.29 $\pm$ 1.41	2.19 $\pm$ 1.27
Soft palate	2.81 $\pm$ 1.26	2.83 $\pm$ 1.42	2.91 $\pm$ 1.19	2.88 $\pm$ 1.15
Tongue	4.58 $\pm$ 1.61	4.06 $\pm$ 1.36	4.41 $\pm$ 1.43	3.62 $\pm$ 1.17
Upper lip	3.42 $\pm$ 1.11	3.89 $\pm$ 1.30	3.15 $\pm$ 1.15	3.21 $\pm$ 1.16
Vocal folds	7.54 $\pm$ 4.20	6.07 $\pm$ 3.64	6.59 $\pm$ 4.20	4.86 $\pm$ 2.87
Lower incisor	3.93 $\pm$ 1.57	2.83 $\pm$ 0.96	3.41 $\pm$ 1.10	2.14* $\pm$ 0.80
Upper incisor	2.56 $\pm$ 0.95	2.90 $\pm$ 1.03	2.32 $\pm$ 0.91	2.40 $\pm$ 0.98
Full10	4.10 $\pm$ 2.56	3.69 $\pm$ 2.15	3.75 $\pm$ 2.39	3.19* $\pm$ 1.85
Full7	3.35 $\pm$ 1.57	3.17 $\pm$ 1.42	3.06 $\pm$ 1.38	2.71* $\pm$ 1.21

Values: mean $\pm$ SD; bold: lowest mean. *Different from A14 (paired two-sided *t*-test, $n = 8$ speakers, unadjusted $p < 0.05$).

Figure 2 illustrates the application of the fixed mapping to a P8 /k/ frame in *avec*, beyond the /u/ calibration vowel.

5. DISCUSSION

These comparisons assess the contributions of anatomical correspondences and nonlinear alignment. A14 improves over A12 by adding two landmarks, whereas A12+T12 improves aggregate performance with the landmark set unchanged. A12+T14 performs best overall, suggesting that the additional lower-incisor correspondences help constrain non-rigid adaptation. Tongue error decreases by 10.8% relative to A14, although this reduction is not statistically significant.

Residual laryngeal errors may reflect anatomical differences, including gender-related oral-pharyngeal proportions and laryngeal position [18, 19], whose contributions are not isolated here. Lip variability and the deterioration for P4 further indicate that benefits are not uniform. A fixed

Table 3. Speaker-wise Full10 P2CP_{mean} (mm). A14 is the main affine baseline; A12+T14 is the proposed configuration.

Speaker	Raw	A12	A14	A12+T12	A12+T14
P1	9.71 $\pm$ 3.63	5.85 $\pm$ 3.41	5.33 $\pm$ 2.94	5.34 $\pm$ 3.51	3.95* $\pm$ 2.31
P3	9.69 $\pm$ 4.26	5.06 $\pm$ 3.38	4.42 $\pm$ 2.49	4.31 $\pm$ 2.89	3.46* $\pm$ 1.76
P4	8.14 $\pm$ 4.26	3.28 $\pm$ 1.64	2.95 $\pm$ 1.36	3.06 $\pm$ 1.93	3.17* $\pm$ 2.10
P5	4.42 $\pm$ 2.12	3.13 $\pm$ 1.36	2.93 $\pm$ 1.21	2.64 $\pm$ 1.03	2.46* $\pm$ 1.00
P6	7.79 $\pm$ 3.37	3.41 $\pm$ 1.97	2.94 $\pm$ 1.58	3.32 $\pm$ 1.69	2.82* $\pm$ 1.36
P7	11.23 $\pm$ 4.55	3.41 $\pm$ 2.27	3.34 $\pm$ 2.26	3.69 $\pm$ 2.15	3.17* $\pm$ 2.14
P8	14.22 $\pm$ 5.27	3.87 $\pm$ 1.99	3.79 $\pm$ 1.84	4.41 $\pm$ 2.00	3.24* $\pm$ 1.44
P9	5.74 $\pm$ 2.01	4.80 $\pm$ 2.14	3.84 $\pm$ 1.67	3.25 $\pm$ 1.85	3.22* $\pm$ 1.89
All	8.87 $\pm$ 4.81	4.10 $\pm$ 2.56	3.69 $\pm$ 2.15	3.75 $\pm$ 2.39	3.19* $\pm$ 1.85

Values: mean $\pm$ SD; bold: lowest mean. *Different from A14 (paired two-sided *t*-test; session means within speakers, speaker-level for All; unadjusted $p < 0.05$).

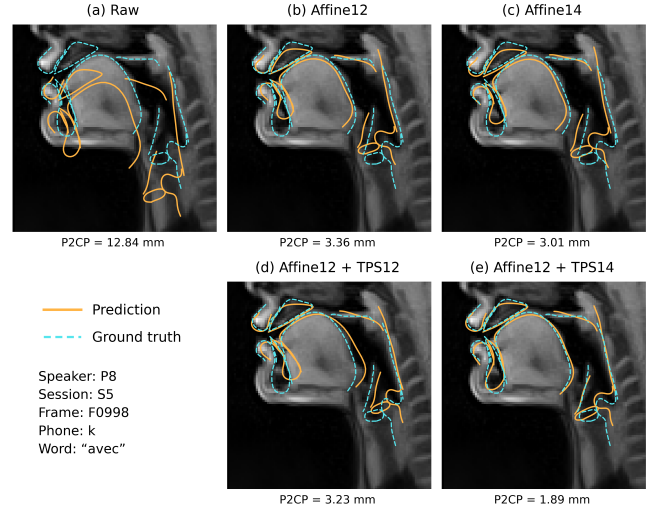


Fig. 2. Adaptation example for P8 containing /k/ in *avec*. Orange solid: predictions; cyan dashed: ground truth. Frame-level P2CP errors are shown below each panel.

mapping may only partially capture speaker-specific articulation [1] and pose differences across unregistered ASD1 recordings[17]. Sensitivity to calibration-frame selection and landmark annotation remains to be quantified.

6. CONCLUSION

Single-frame anatomical adaptation transfers vocal-tract contours from a fixed AAI model to unseen speakers without retraining. Across eight speakers, Affine12+TPS14 reduces Full10 error from 3.69 mm with Affine14 to 3.19 mm, supporting landmark coverage and nonlinear alignment. Future work will extend this framework to multi-speaker training toward speaker-independent AAI, while assessing calibration robustness and residual laryngeal and lip errors.

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